

KUMARI SHIVANGI GUPTA

Adm. No. 24MT0216

✉ 24mt0216@iitism.ac.in

☎ +91-7488257838

🌐 [linkedin.com/in/shivangigupta](https://www.linkedin.com/in/shivangigupta)



Summary

Enthusiastic and detail-oriented engineering graduate with solid foundation in technical principles, problem solving, and teamwork. Eager To apply academic knowledge and hands-on project experience to real-world engineering challenges in a professional setting. Committed to continuous learning and innovation.

Education

Indian Institute of Technology

Expected May 2026

Master of technology in Electronics Engineering (CGPA: 7.78/10)

Dhanbad, Jharkhand

- Specialization: RF & Microwave Engineering – This field combines electromagnetic theory, circuit design, signal processing, and materials science to enable efficient transmission and reception of high frequency signals.

Chaibasa Engineering College

2017-2021

Bachelor of technology in Electronics & Communication Engineering (CGPA: 8.15/10)

Chaibasa, Jharkhand

- Electronics and Communication Engineering – This field is dynamic branch of engineering that integrates electronic device design with communication technologies, this field is integral to the development of smart systems.

Internship

National Institute of Technology

Dec 2019

Project title

Jamshedpur, Jharkhand

- **IOT (Proteus, Keil, Arduino), EDA Tool (Microwind), Image Processing** – EDA tools are software applications used to design, simulate, verify, and manufacture electronic systems such as ICs, PCBs, and embedded systems, Keil is a professional development tool suite used primarily for embedded systems programming

Indian Railways

April 2021

Project title

Chakradharpur, Jharkhand

- **Signal & Telecom, Optical Fiber & networking** – In modern rail networks (like metros or high- speed rail), all three- optical fiber, signaling, and telecom work together as part of integrated railway communication and control systems to enhance safety, efficiency, real-time monitoring and automation.

Projects

- **Project Name (final year project of M. Tech) / Rotation sensors based on metamaterial**
Currently designing and simulating a Rotation Sensors based on metamaterial to measure angle, direction, or speed of rotation of an object like shaft, wheel or motor using Ansys HFSS and CST studio.
- **Project Name / Dielectric Resonator Antenna (DRA) Array**
In this project, a high gain 1*2 Dielectric Resonator Antenna (DRA) Array designed and simulated using ANSYS HFSS for 5G-Millimeter-Wave applications.
- **Project Name / Design and simulation of LNA on ADS**
A LNA with 3 Watts transistor at 700MHz with Gain greater than 15 dB and Noise Figure less than 0.5 dB is design using ADS.
- **Project Name / 8-bit Ripple carry adder**
An 8-bit Ripple carry adder consists of 8 full adders chained together. Each full adder handles one bit from two 8-bit inputs and passes its carry output to the next adder stage.
- **Project Name (final year project of B. Tech) / Biometric attendance system using Arduino**
A biometric attendance system using Arduino uses fingerprint recognition to automatically and securely track attendance, it is microcontroller-based project and Arduino platform uses C++ language to store student names.

Technical Skills

- Tools and framework- Ansys HFSS, Keysight ADS, CST studio, Proteus, Microwind, Keil, MS Office, Arduino, Raspberry Pi.
- Programming Language- Python, Verilog HDL, MATLAB, Machine Learning beginner level.
- Operating Systems- Windows & Linux

Subjects of interest

- Analog circuits
- Digital circuits
- Electromagnetic theory
- Electronics devices & circuits
- Antenna theory

Achievements

- Gate Qualified 2024 in Electronics and Communication Engineering
- Certificate of Internships from Indian Railways and NIT Jamshedpur in April 2021 and Dec 2019.
- Paper published on IEEE MAPCON 2025 conference entitled as “A novel modified Hryvnian Shaped Resonator Loaded Microwave Rotation Sensor for Bio-Medical Motion Tracking and Monitoring”.
- Attended MAPCON 2025 conference.